

SONiX MCU CRC Tool

User Guide

Version 1.0

SN8F5830

SN32F280/290

SN32F780/790

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AMENDENT HISTORY

Version	Date	Description
1.0	2020/05/08	First version is released.

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1 OVERVIEW

为使 SONiX MCU 能支持 CRC Calculator 功能，故提供了 CRC Calculator Tool，并结合 Keil 编译环境一起使用，以下将说明环境设置与程序编译流程。

1.1 CRC Calculator Tool 特点

- 支持 8051 与 M0 系列 MCU.
- 依照不同 IC 型号自动计算 ROM CRC checksum.
- 可提供客户方便验证 CRC 结果

2 环境设置

User 务必先按照下列方式设置好环境，才可执行后续流程。

2.1 8051 环境设置

- (1). 前往 SONiX 官网(<http://www.sonix.com.tw/>)下载 SN-Link Driver for C51.
- (2). 安装 SN-Link_Driver for C51_V2.00.37.exe
 - 请务必确认是 V2.00.37 或以后版本.
- (3). 此工具会自动安装到 Keil 的安装路径下

2.2 M0 环境设置

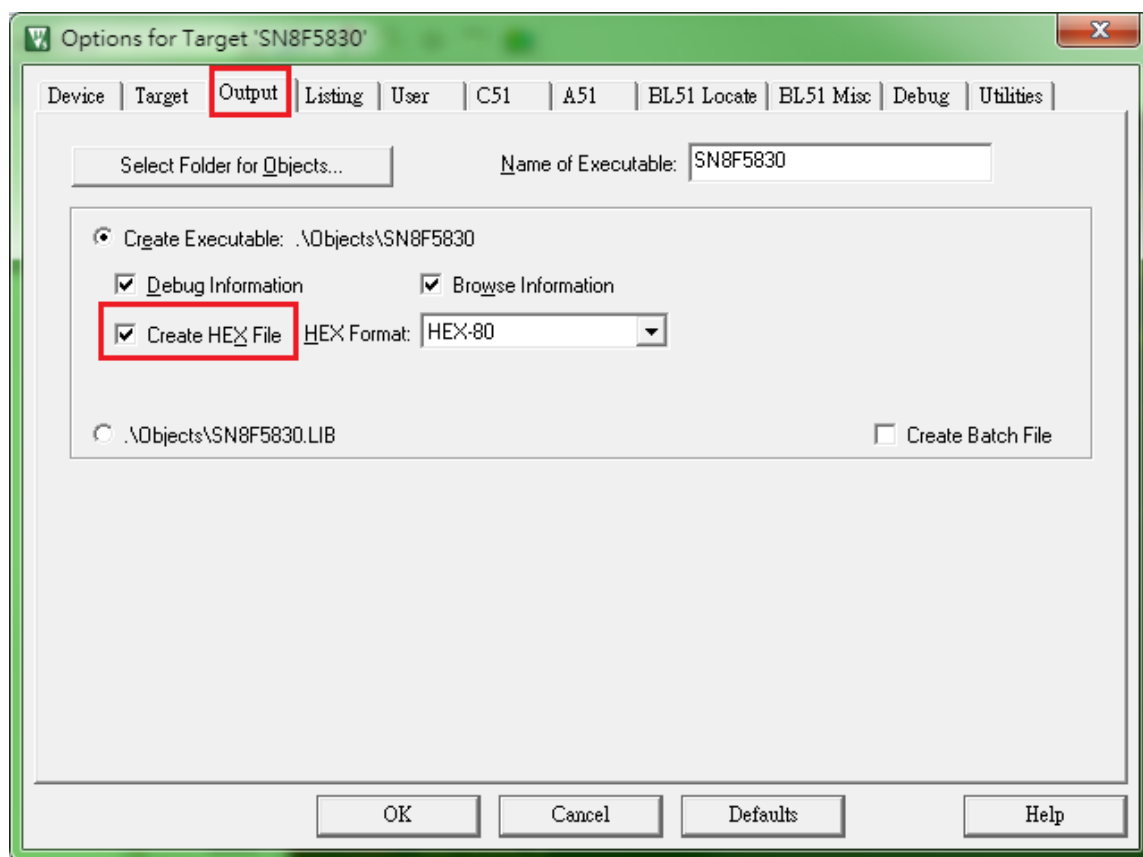
- (1). 前往 SONiX 官网(<http://www.sonix.com.tw/>)下载 SN-Link Driver for M0.
- (2). 安装 SN-Link_Driver for Keil Cortex M0_V2.00.319.exe
 - 请务必确认是 V2.00.319 或以后版本.
- (3). 此工具会自动安装到 Keil 的安装路径下

3 产生 CRC 结果

3.1 8051 环境设置 (以 SN8F5835 为例)

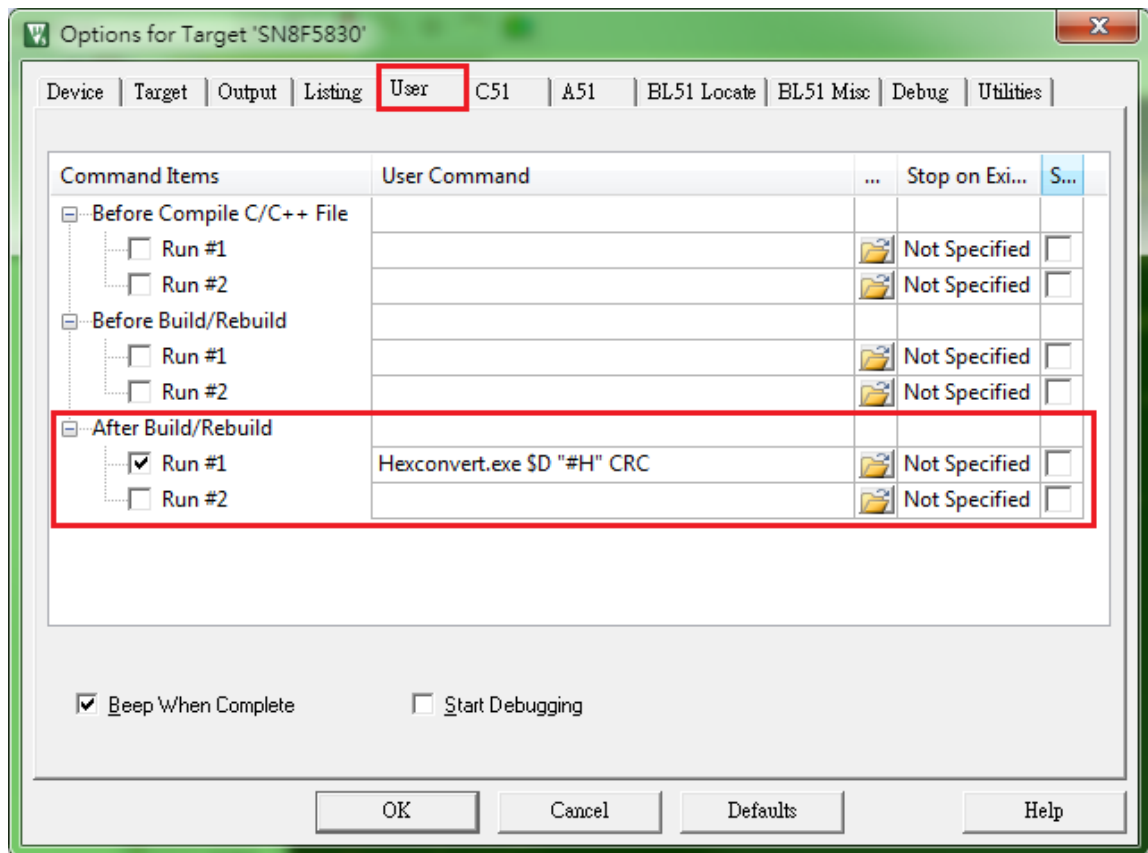
➤ 点选 Keil 界面的 Option of Target > Output, 如下图。

1. 确认勾选 Create HEX File.



➤ 点选 Keil 界面的 Option of Target > User，如下图。

1. 确认勾选 Run #1，并输入 HexConvert.exe \$D "#H" CRC.



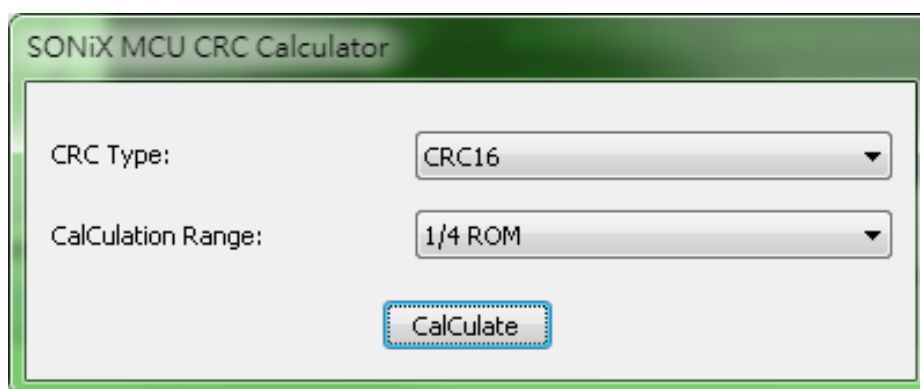
- 填写完成后，执行 **compiler**。



Build(F7)

- **Keil** 会在 **compiler** 完成之后，弹跳出 **SONiX MCU CRC Calculator.exe** 执行档，如下图。

1. 请选择要计算的 CRC Type.
 - CRC16
 - CRC16-CCITT
2. 请选择要计算的 Calculation Range.
 - 1/4 ROM
 - 1/2 ROM
 - Avoid the last 8 page
 - Avoid the last 1 page + 2 Byte



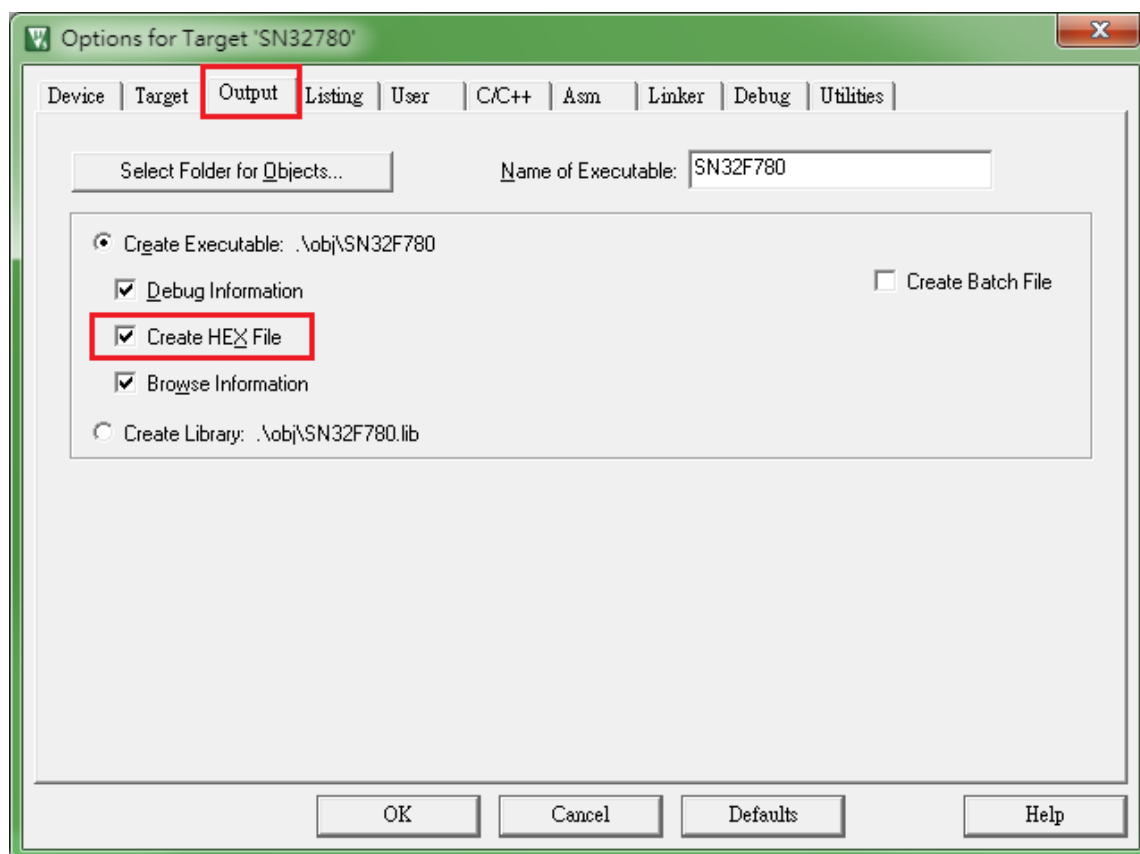
- 按下 **Calculate** 按钮后，AP 会计算出 **CRC** 结果并将该信息显示在 **Keil** 的 **Build Output window** 上。

```
CRC_Type = CRC16
CRC_CalculationRange = 1/4 ROM
CRC_Checksum = 0xB7D7
```

3.2 M0 环境设置 (以 SN32F780 为例)

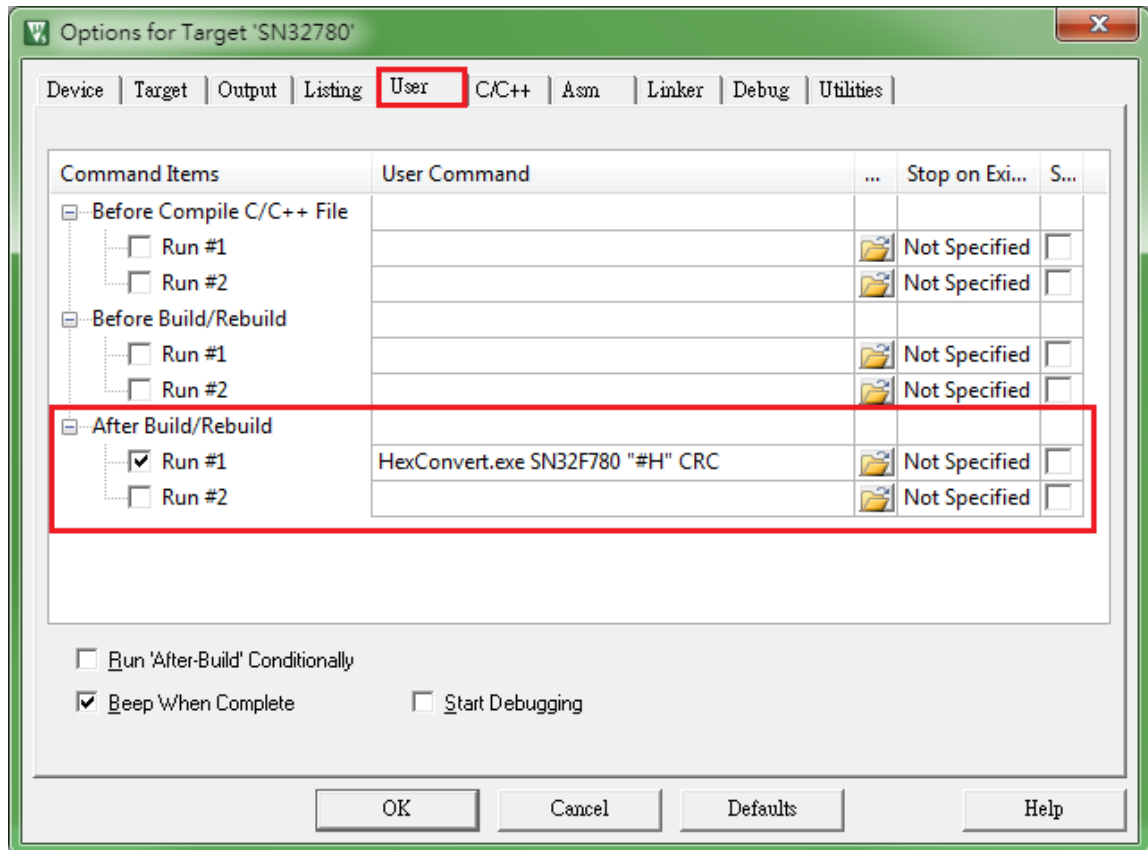
➤ 点选 Keil 界面的 Option of Target > Output, 如下图。

1. 确认勾选 Create HEX File.

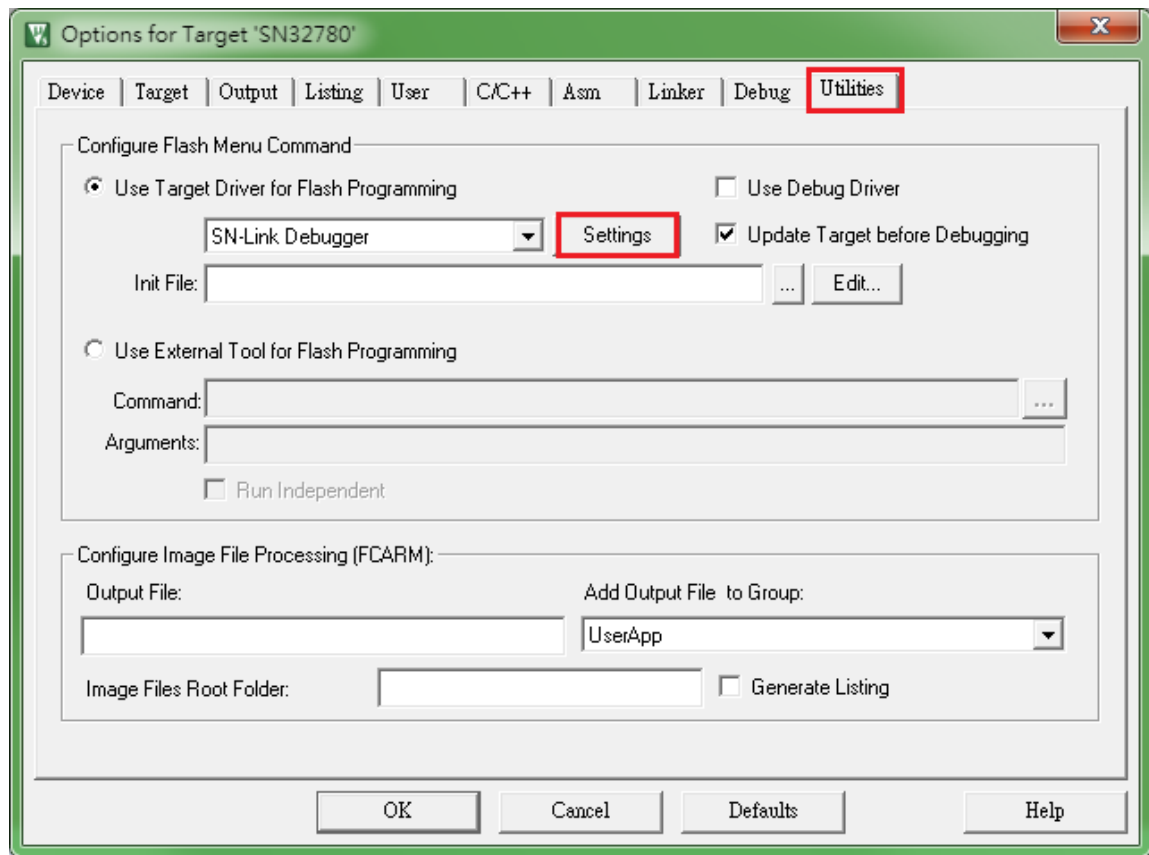


➤ 点选 Keil 界面的 Option of Target > User，如下图。

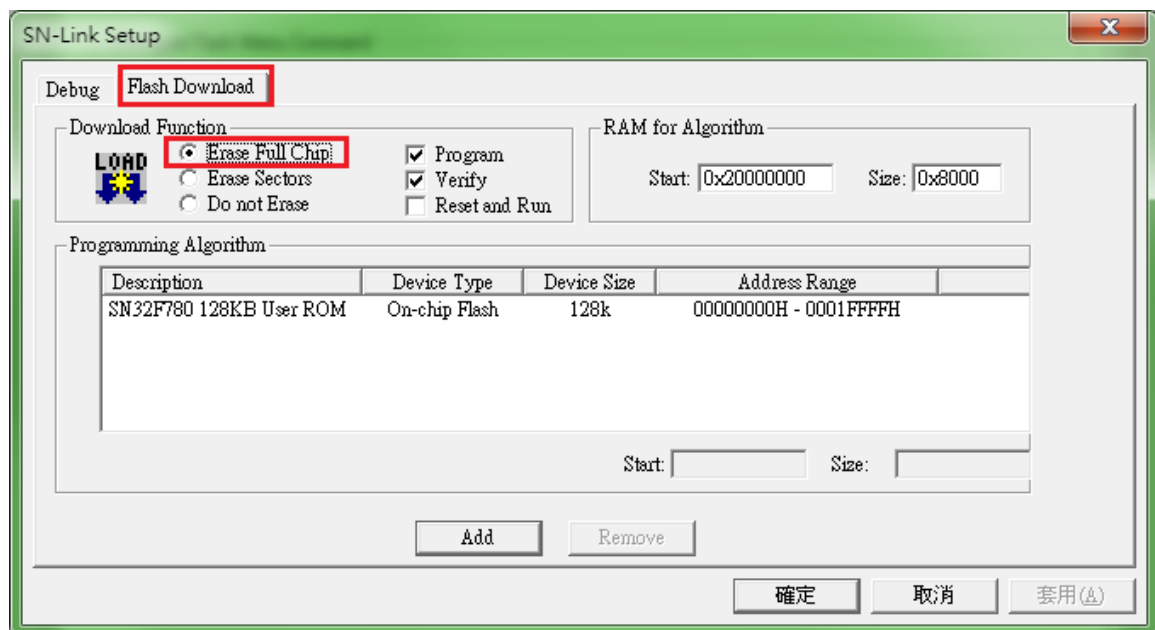
1. 确认勾选 Run #1，并输入 HexConvert.exe SN32F780 "#H" CRC.
2. 其中 SN32F780 依照各 chip name 修改.



- 点选 Keil 界面的 Option of Target > Utilities > 点击 Settings，如下图。



- Download Function 中选择 Erase Full Chip，如下图。



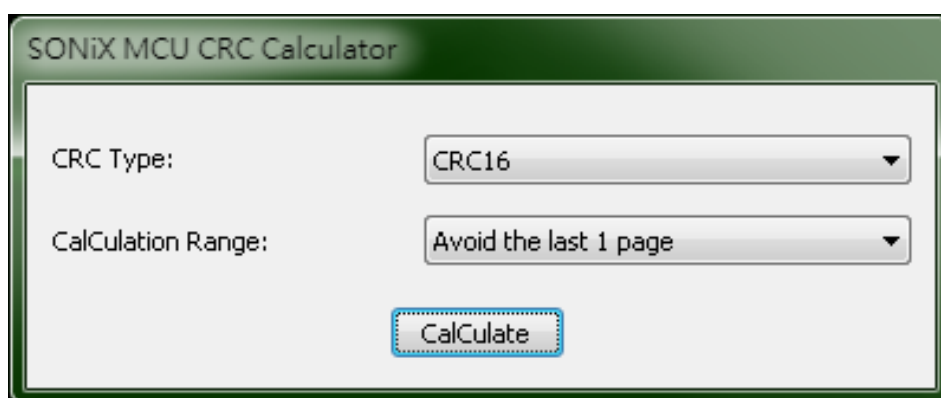
- 填写完成后，执行 **compiler**。



Build(F7)

- **Keil** 会在 **compiler** 完成之后，弹跳出 **SONiX MCU CRC Calculator.exe** 执行档，如下图。

1. 请选择要计算的 CRC Type.
 - CRC16
 - CRC16-CCITT
 - CRC32
2. 请选择要计算的 Calculation Range.
 - Avoid the last 1 Page



- 按下 **Calculate** 按钮后，AP 会计算出 **CRC** 结果并将该信息显示在 **Keil** 的 **Build Output window** 上。

```
CRC_Type = CRC16
CRC_CalculationRange = Avoid the last 1 page
CRC_Checksum = 0x61B4
```

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